# Water groups and arrays

H2O: Why Water Is Special

Name _____ Date _____

Write one answer in each blank. Use repeated addition to show every equal group or row.

**1.** Big fills 2 ice trays with 4 ice cubes in each tray. How many ice cubes are there altogether?

$4 + 4 =$ _____ Show your thinking.

**2.** Little makes 3 rows of 3 water drops on wax paper. How many drops are in the array?

$3 + 3 + 3 =$ _____ Show your thinking.

**3.** There are 5 cups with 2 paper straws beside each cup. How many paper straws are there altogether?

$2 + 2 + 2 + 2 + 2 =$ _____ Show your thinking.

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**4.** Big arranges 3 rows of 5 paperclips for a water test. How many paperclips are in the array?

$5 + 5 + 5 =$ _____

Show your thinking.

**5.** Little sets 4 trays with 4 small bowls of water on each tray. How many bowls are there altogether?

$4 + 4 + 4 + 4 =$ _____

Show your thinking.

**6.** Big arranges 5 rows of 4 cups of water. How many cups are in the array?

$4 + 4 + 4 + 4 + 4 =$ _____

Show your thinking.

## Three atoms

Every bit of water is made of molecules. Each H2O molecule has two hydrogen atoms joined to one oxygen atom.





## Three states

The same water can be solid ice, flowing liquid, or invisible water vapor. Melting and freezing change its state.





## Drops gather

Water molecules attract one another. At the surface, that pull gathers drops into rounded beads that can join together.





## Surface tension

Surface tension makes the top of water resist being pulled apart. A gently placed paperclip can rest there without sinking.





## Roomier ice

Ice molecules form a roomier pattern. Ice is lighter for its size than liquid water, so it floats on top.





## Water dissolves

Water can surround and carry away tiny pieces of salt or sugar. That is called dissolving, but not everything dissolves.



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!





WONDER DAILY

# H2O: Why Water Is Special

# Write what water can do

H2O: Why Water Is Special · Curiosity day

Name _____ Date _____

Choose a clue. Say your idea first, then draw or write it.



Three atoms together



Drops pull together



Ice floats high

Draw one water clue and add a small arrow to the part you want a reader to notice.

water · molecule · atom · hydrogen · oxygen · ice · melt · freeze · vapor · surface tension · float · dissolve

One water molecule has … dictate the number and kinds of atoms

A drop beads up because … dictate what the molecules do

Ice floats because … dictate how ice compares with liquid water

Does your sentence name a water clue and explain why it happens?